

Problem 1: -Design and develop a reprogrammable embedded computer using 8051 Microcontroller.

Solution:

Source Code:

```
#include<reg51.H>

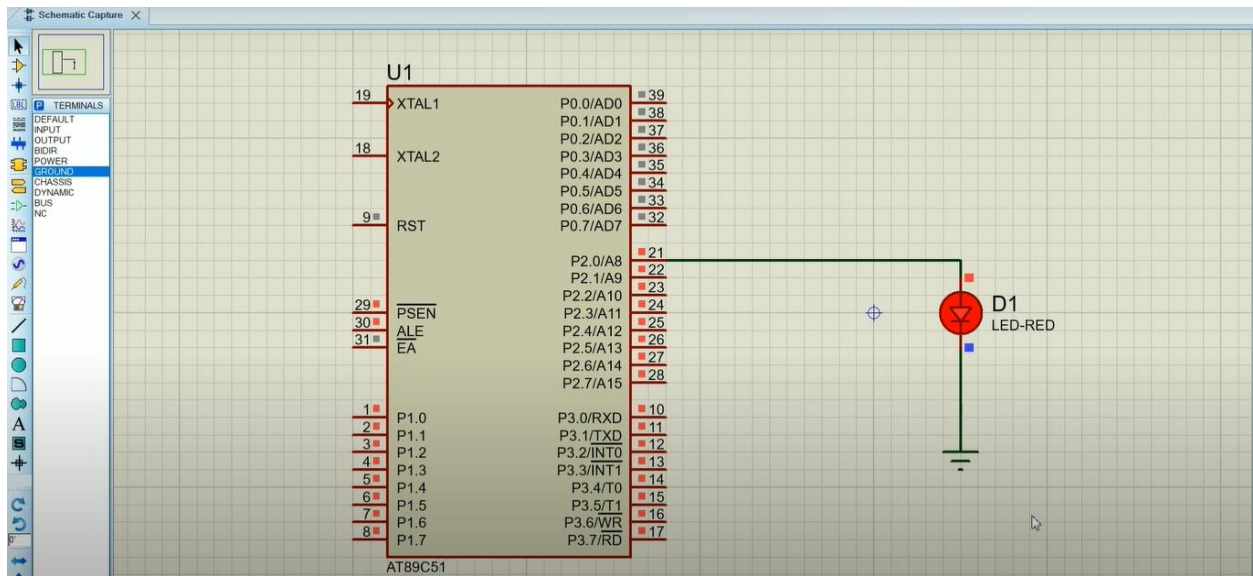
sbit LED = P2^0;

void Delay(void);

void main(void)
{
    while(1)
    {
        LED = 0;
        Delay();
        LED = 1;
        Delay();
    }
}

void Delay(void)
{
    int j;
    int i;
    for(i = 0; i < 10; i++)
    {
        for(j = 0; j < 10000; j++)
        {
        }
    }
}
```

Output:



Problem 2(a): - Configure Timer control registers of 8051 and develop a program to generate delay

Solution:

Source code:

```
#include<reg51.H>
```

```
void Delay(void);
```

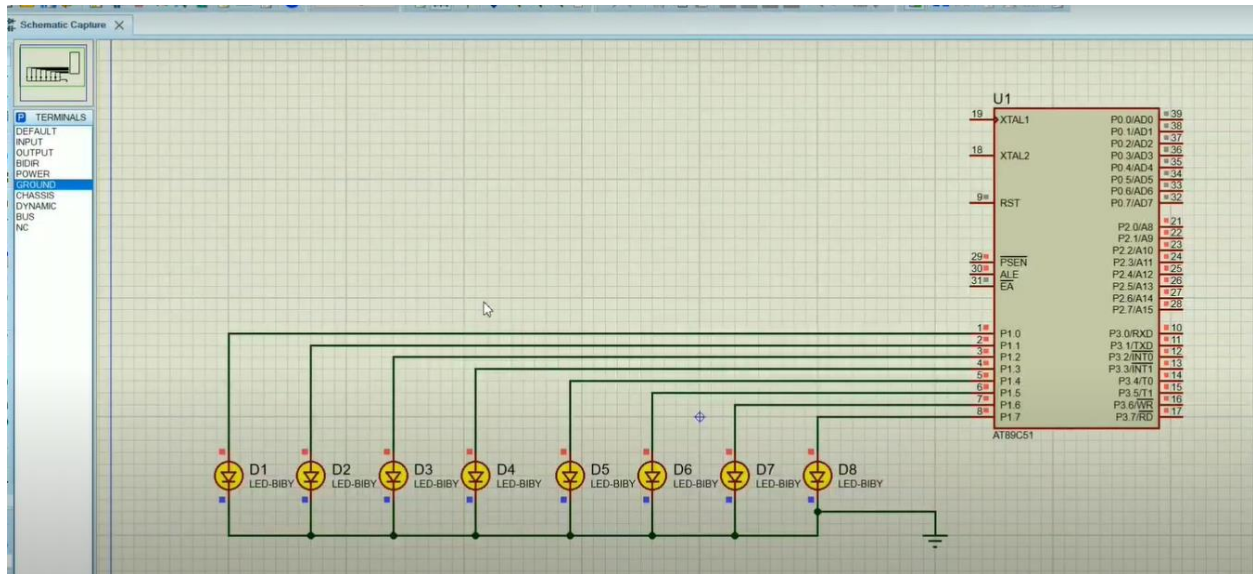
```
void main(void)
```

```
{  
    while(1)  
    {  
        P1 = 0xFF;    // Make all bits of P1 high  
        Delay();  
        P1 = 0x00;    // Make all bits of P1 low  
        Delay();  
    }  
}
```

```
void Delay(void)
```

```
{  
    int j;  
    int i;  
    for(i = 0; i < 1000; i++)  
    {  
    }  
    for(j = 0; j < 1000; j++)  
    {  
    }  
}
```

Output:



Problem 2(b): - Use of general purpose I/O port of two controllers for data transfer.

Solution:

Source code:

Controller 1-

```
#include<reg51.h>

void Delay(unsigned int time);

void main(void)
{
    P2 = 0xAA;
    Delay(1000);
    P2 = 0x55;
    Delay(1000);
}

void Delay(unsigned int time)
{
    unsigned int i,j;
    for (i = 0; i < time; i++){
        for (j = 0; j < 23; j++){
            }
        }
    }
}
```

Controller 2-

```
#include<reg51.h>

void Delay(unsigned int time);

void main(void)
{
    while(1){
```

```

        P2 = P1;

        Delay(1000);

    }

}

void Delay(unsigned int time)

{

    unsigned int i,j;

    for (i = 0; i < time; i++){

        for (j = 0; j < 23; j++){

            }

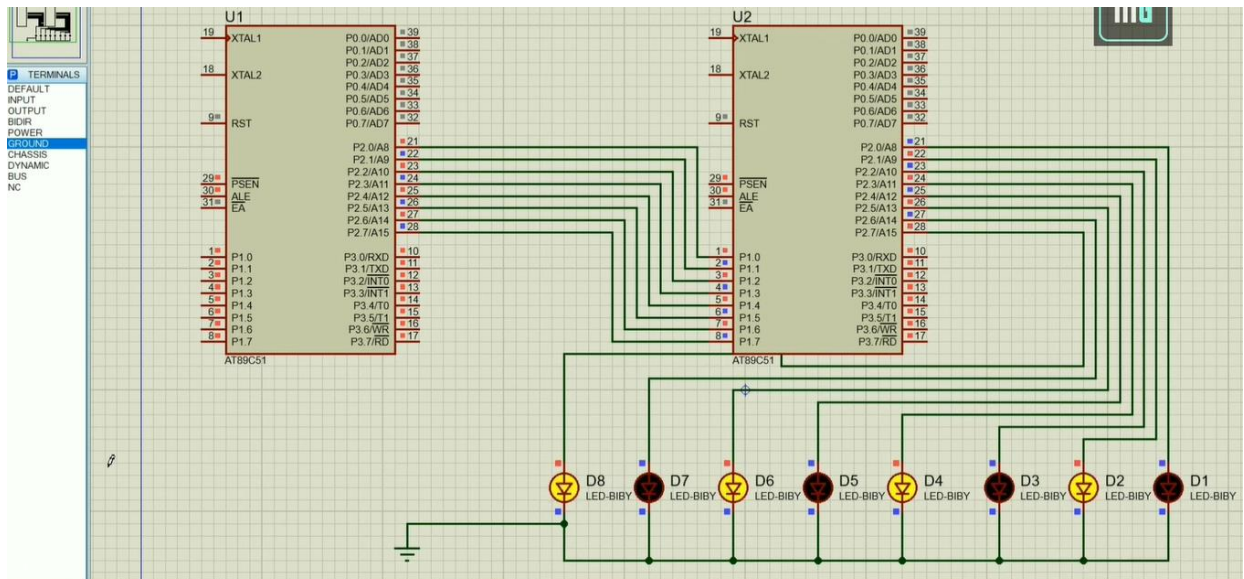
        }

    }

}

```

Output:



Problem 3(a): Simulate Binary counter (8 bits) on LEDs.

Solution:

Source Code:

```
#include<reg51.h>

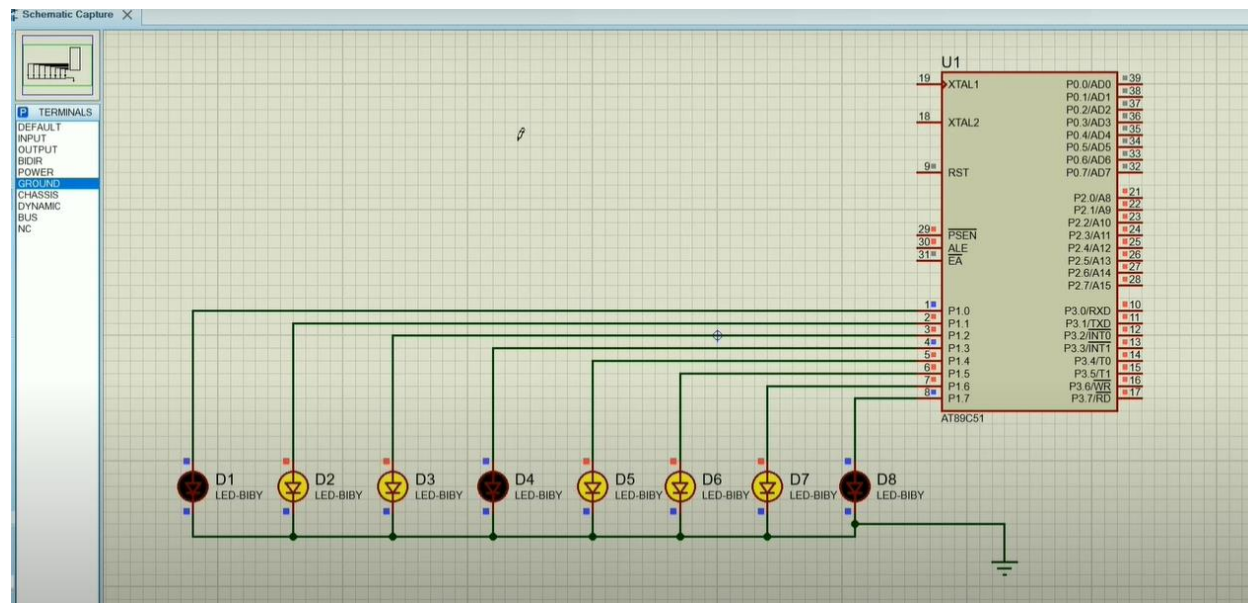
void delay(int time);

void main()
{
    P1 = 00000000;
    while(1)
    {
        P1++;
        delay(100);
    }
}

void delay(int time)
{
    int i, j;
    for(i = 0; i <= time; i++)
    {
        for(j = 0; j <= 23; j++)
        {
        }
    }
}
```

The screenshot shows a Proteus 8.0 Schematic Capture window. The circuit diagram features an 8051 microcontroller (U1, AT89C51) connected to eight LEDs (D1-D8). The LEDs are arranged in two rows of four. The P0 port of the microcontroller is connected to D1, D2, D3, and D4. The P1 port is connected to D5, D6, D7, and D8. The P2 and P3 ports are not connected to any LEDs. The circuit is powered by a 5V supply and ground. The terminal list on the right shows the pin connections for U1.

Pin	Signal	Value
19	XTAL1	=39
18	XTAL2	=38
9	RST	=37
29	PSEN	=36
30	ALE	=35
31	EA	=34
18	P1.0	=21
2	P1.1	=22
3	P1.2	=23
4	P1.3	=24
5	P1.4	=25
6	P1.5	=26
7	P1.6	=27
8	P1.7	=28
10	P3.0/RXD	=10
11	P3.1/TXD	=11
12	P3.2/INT0	=12
13	P3.3/INT1	=13
14	P3.4/T0	=14
15	P3.5/T1	=15
16	P3.6/T2	=16
17	P3.7/RD	=17



Problem 3(a): To interface 8 LED's at I/O ports and create different patterns

Solution:

Source Code:

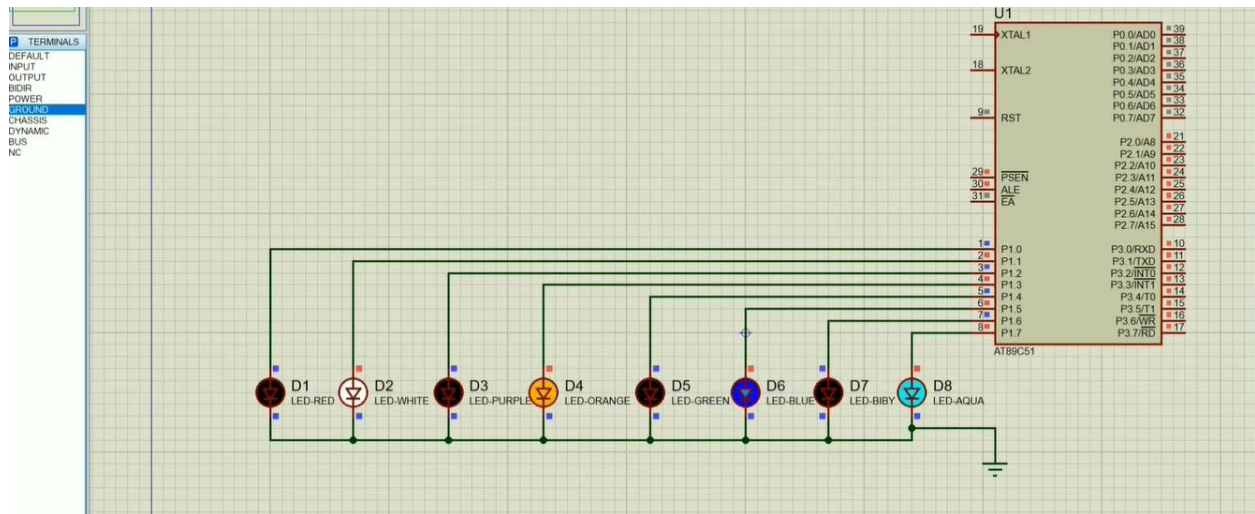
```
#include<reg51.h>

void delay();

void main()
{
    while(1)
    {
        P1 = 0xAA; // Pattern to turn on alternate LEDs - 10101010
        delay();
        P1 = 0x55; // Reverse the pattern
        delay();
    }
}

void delay()
{
    unsigned int i, j;
    for(i = 0; i < 23; i++)
    {
        for(j = 0; j < 1000; j++)
        {
        }
    }
}
```

Output:



Problem 4(b): To Demonstrate Seven Segment LED Display and Generate counting from 0 to 99

Solution:

Source Code:

```
#include<reg51.h>
```

```
void delay(unsigned int ms)
```

```
{
```

```
    unsigned int i, j;
```

```
    for(i = 0; i<ms; i++)
```

```
    {
```

```
        for(j = 0; j<=1275; j++)
```

```
        {
```

```
        }
```

```
    }
```

```
}
```

```
void main(void)
```

```
{
```

```
    char number[] = {0x3F, 0x06, 0x5B, 0x4F, 0x66, 0x6D, 0x7D, 0x07, 0x7F, 0x6F};
```

```
    int i, j;
```

```
    P2 = 0x00;
```

```
    P3 = 0x00;
```

```
    while(1)
```

```
    {
```

```
        for(i = 0; i<=9; i++)
```

```
        {
```

```
            P2 = number[i];
```

```
            for(j = 0; j<=9; j++)
```

```
            {
```

```

        P3 = number[j];

        delay(50);

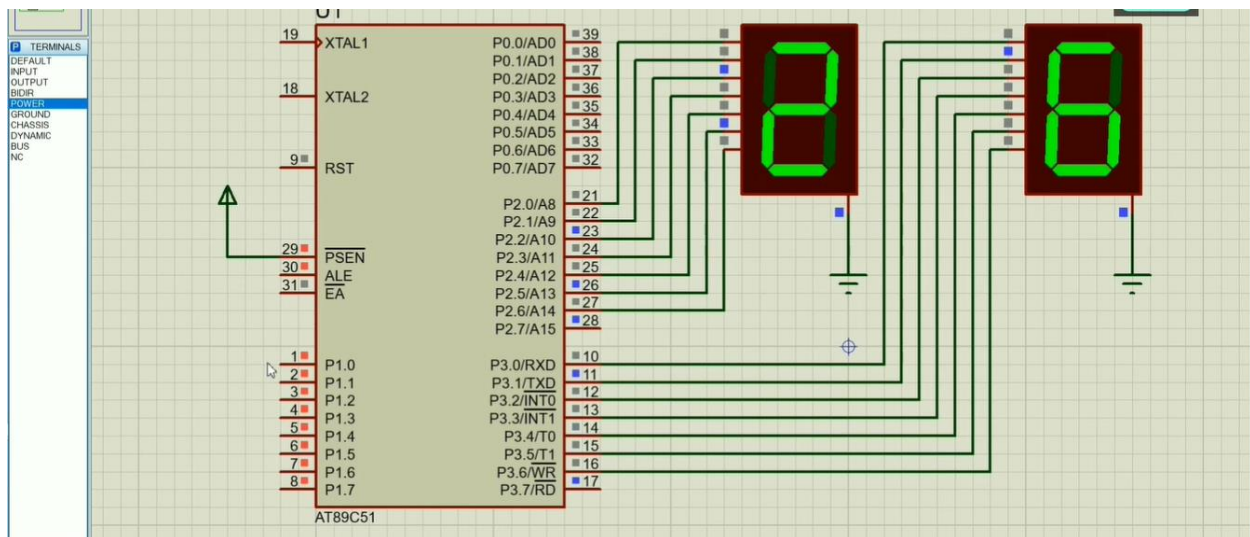
    }

}

}

```

Output:



Problem 5: Develop a Program for interfacing stepper motor with 8051

Solution:

Source Code:

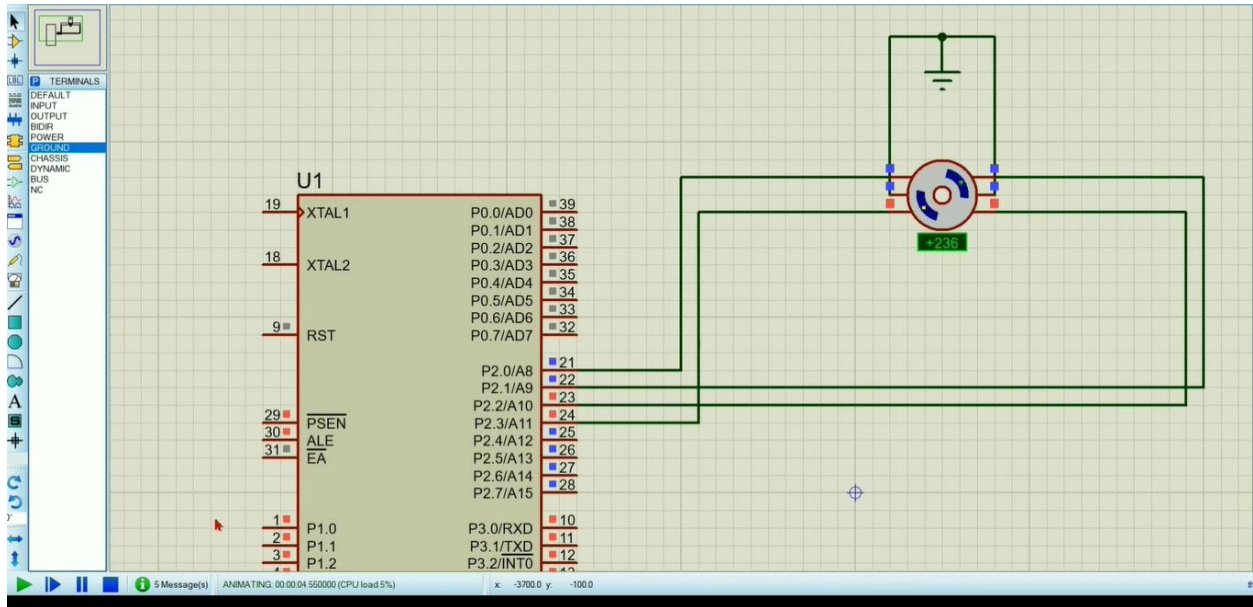
```
#include<reg51.h>

void delay()
{
    int i, j;
    for(i = 0; i <= 100; i++)
    {
        for(j = 0; j < 100; j++)
        {
        }
    }
}

void main()
{
    while(1)
    {
        P2 = 0x09;
        delay();
        P2 = 0x03;
        delay();
        P2 = 0x06;
        delay();
        P2 = 0x0C;
        delay();
    }
}
```

}

Output:



Problem 6: Generate Traffic Light Signal

Solution:

Source Code:

```
#include<reg51.h>

//signal 1
sbit tl1r = P2^0;    // red
sbit tl1o = P2^1;    // orange
sbit tl1g = P2^2;    // green

//signal 2
sbit tl2r = P2^3;    // red
sbit tl2o = P2^4;    // orange
sbit tl2g = P2^5;    // green

//signal 3
sbit tl3r = P3^0;    // red
sbit tl3o = P3^1;    // orange
sbit tl3g = P3^2;    // green

void delay(int t);
void trafficlight(void);

void main()                // main program
{
    P1=0x00;
    P2=0x00;
    P3=0x00;

    while(1)
    {
        trafficlight();
    }
}

void delay(unsigned long int t)    // delay routine
{
    while(t>0)
    {
        unsigned long int i;
        for(i=1;i<10*1275;i++);
        t--;
    }
}
```

```

}

void trafficlight(void)          // traffic light system program

{

P2= 0x11;  // traffic signal control
P3= 0x04;


t11r=1;    // signal 1
t11o=0;
t11g=0;


t12r=0;    // signal 2
t12o=1;
t12g=0;


t13r=0;    // signal 3
t13o=0;
t13g=1;


delay(100); // delay

P2= 0x0c;   // traffic signal control
P3= 0x02;


t11r=0;     // signal 1
t11o=0;
t11g=1;


t12r=1;     // signal 2
t12o=0;
t12g=0;


t13r=0;     // signal 3
t13o=1;
t13g=0;


delay(100); // delay

P2= 0x22;   // traffic signal control
P3= 0x01;


t11r=0;     // signal 1
t11o=1;
t11g=0;


t12r=0;     // signal 2

```

```

t12o=0;
t12g=1;

t13r=1;    // signal 3
t13o=0;
t13g=0;

delay(100); // delay
}

```

Output:

